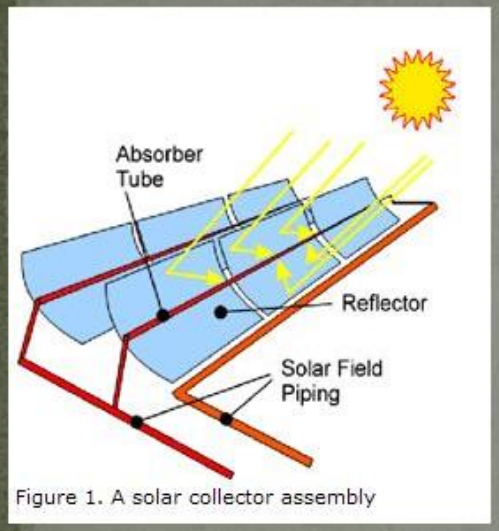


Figure 1. A solar collector assembly





http://www.nrel.gov





Dr. Hong Nam Nguyen
General Education Department
nguyen-hong.nam@usth.edu.vn

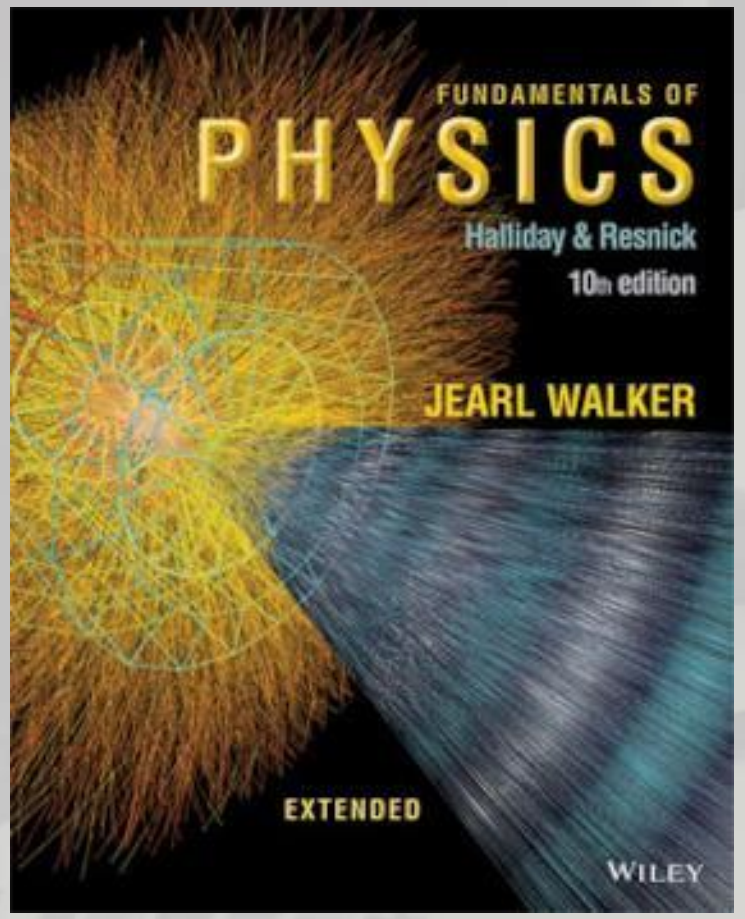
Table of Contents

- Overview and basic concept of thermodynamics
- Temperature, Heat, and the First Law of Thermodynamics
- The kinetics theory of gases
- Second law of Thermodynamics and applications

Evaluation

- Attendance: 10% (final exam prohibited for those absent >30%)
- Mid-term exam + participation in tutorial classes (30%)
- Final exam (60%)

I. Overview and basic concepts



Outcomes

- Remember the historical development of thermodynamics as a branch of science
- Understand some terminologies in thermodynamics
- Understand some basic concepts in thermodynamics

I. Overview and basic concepts



Thermodynamics

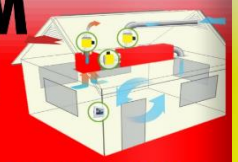
- Heat (thermo) + movement (dynamics)
- A branch of science which deals with energy transfer and its effects on properties (physical and chemical) of the substance
- Lots of applications in daily life and industries are engineered using the principles of thermodynamics

I. Overview and basic concepts

NOTE: *HVAC stands for heating, ventilation, and air conditioning*

HVAC SYSTEM

FOURTH APPLICATION

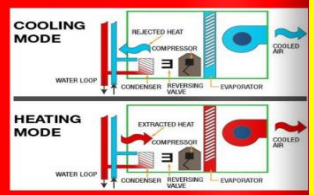


THEORY & LAW INVOLVED

- **First Law of Thermodynamics**
  Although heat energy can't be created and destroyed, heat energy can be moved from one location to another when we want to heat or cool the building.
- **Second Law of Thermodynamics**
  100% efficiency of energy conversion does not exist to keep building warm such as heat pumps.
- **Specific heat**
  Less work required to keep water warm in water-heated radiant floors because water has higher specific heat capacity.
- **Conduction**
  Building is insulated to trap the heat.
- **Convection**
  Employs a fan to force the heated air throughout the building, creating circulation pattern.
- **Radiation**
  Heat lamps will transfer heat energy.

WORKING PRINCIPLE

- **During cold weather (Heating)**
  1) Evaporator absorbs heat from outside air.
  2) Compressor compresses refrigerant gas and temperature increases.
  3) Condenser uses fans to cool refrigerant gas and heat is released to inside.
- **During hot weather (Cooling)**
  1) Evaporator absorbs heat from inside air.
  2) Compressor compresses refrigerant gas and temperature increases.
  3) Condenser uses fans to cool refrigerant gas and heat is released to outside.



HVAC system applies laws of thermodynamics to heat up or cool down the building.

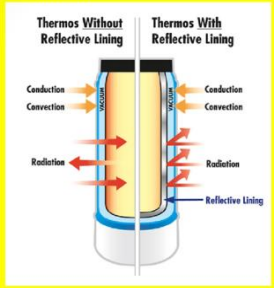
VACUUM FLASK

SECOND APPLICATION



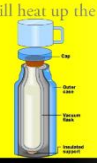
THEORY & LAW INVOLVED

- **First Law of Thermodynamics**
  If vacuum flask is isolated system, then no heat is created and destroyed to heat up or cool down the liquid.
- **Conduction**
  Vacuum between walls stop heat transfer by conduction.
- **Convection**
  Plastic stopper stops heat transfer by convection.
- **Radiation**
  Silver coating stops heat transfer by radiation.



WORKING PRINCIPLE

- **Keeping things hot**
  1) Vacuum between walls stop heat leaving from vacuum flask by conduction.
  2) Plastic stopper stops heat rising away by convection in the form of vapor.
  3) Shiny surface inside the vacuum flask reflects thermal radiation back to its content.
- **Keeping things cold**
  1) Vacuum between walls stop heat entering vacuum flask by conduction.
  2) Plastic stopper stops heat entering the flask by convection.
  3) Shiny surface outside the vacuum flask reflect thermal radiation away which will heat up the thermal flask.



Vacuum flask applies laws of thermodynamics to maintain temperature of its contents

MICROWAVE OVEN

THIRD APPLICATION

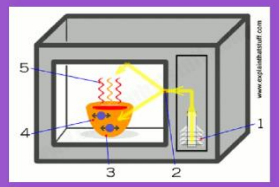


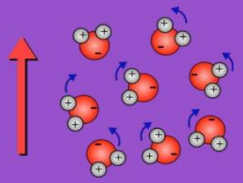
THEORY & LAW INVOLVED

- **Zeroth Law of Thermodynamics**
  Microwave oven spins food slowly to cook food evenly and thermal equilibrium can be reached.
- **First Law of Thermodynamics**
  Radiant energy is converted into thermal energy which is used to cook food.
- **Conduction**
  Heat energy is transmitted through collisions between water molecules.
- **Convection**
  A fan is employed to circulate the hot air to cook the food.
- **Radiation**
  Microwave radiation is emitted to agitate the molecules of food which produces heat energy without physical contact.

WORKING PRINCIPLE

- **How microwave oven cooks food?**
  1) Magnetron generates microwave by using electricity.
  2) Magnetron blasts these waves into the food compartment through a channel called a wave guide.
  3) Turntable is spinning slowly round so the microwaves can cook the food evenly.
  4) Microwaves bounce back and forth off, penetrating the molecules to vibrate them.
  5) Vibration of molecules produces heat energy to cook the food.





Microwave oven applies laws of thermodynamics to heat and cook the food

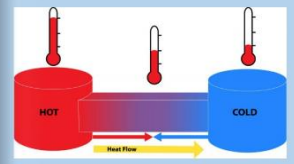
THERMOMETER

FIRST APPLICATION



THEORY & LAW INVOLVED

- **Zeroth Law of Thermodynamics**
  Temperature of thermometer indicates the temperature of certain object when thermometer reaches thermal equilibrium with that object.
- **First Law of Thermodynamics**
  Heat energy is converted into kinetic energy which will cause expansion of mercury.
- **Second Law of Thermodynamics**
  Heat will flow from thermometer to object or object to thermometer depending on their temperature.
- **Thermal expansion**
  Mercury will expand when heated because the molecules move faster and take more space.



WORKING PRINCIPLE

- **When temperature of thermometer is greater than temperature of object**
  1) When thermometer and object in contact with each other, heat energy flows from object to thermometer.
  2) Heat energy is converted into kinetic energy gained by mercury molecules in thermometer, causing expansion of thermometer.
  3) This process continues until thermal equilibrium is reached when temperature of thermometer is same as temperature of object.
- **When temperature of thermometer is lower than temperature of object**
  1) When thermometer and object in contact with each other, heat energy flows from thermometer to object.
  2) Heat energy is lost as kinetic energy by mercury molecules in thermometer, causing contraction.
  3) This process continues until thermal equilibrium is reached when temperature of thermometer is same as temperature of object.

Thermometer applies laws of thermodynamics to measure the temperature of object.

I. Overview and basic concepts

AUTOMOTIVE INDUSTRY (HEAT ENGINE)

SEVENTH APPLICATION

THERMODYNAMICS CYCLE

• Carnot Cycle (Power cycle)

THEORY & LAW INVOLVED

• **First Law of Thermodynamics**
Heat energy can be converted into mechanical energy, no heat engine can have thermal efficiency greater than 100 percent.

• **Second Law of Thermodynamics**
Heat energy cannot be fully converted into mechanical energy, no heat engine can have 100 percent thermal efficiency.

• **Isothermal process**
Heat is absorbed and released by gas to do work on or by the system, causing piston moving up and down.

• **Adiabatic process**
Gas is expanded and compressed to do on or by the thermally insulated system, causing the piston moving up and down.

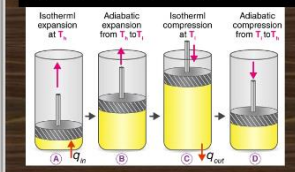
WORKING PRINCIPLE

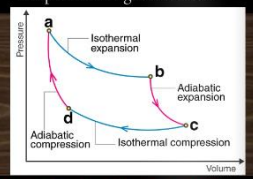
• **Isothermal expansion**
1) Gas absorbs heat at a constant temperature.
2) Gas does work on the system and expands.

• **Adiabatic expansion**
1) Gas continues expanding without heat transfer
2) Gas does work on the system and temperature of gas decreases.

• **Isothermal compression**
1) Gas releases heat a constant temperature
2) **Work done by system and gas is compressed**

• **Adiabatic compression**
1) Gas continues compressing without heat transfer.
2) Work done by system and temperature of gas increases.





Heat engine applies laws of thermodynamics to make car moves.

AEROSPACE INDUSTRY (JET ENGINE)

EIGHTH APPLICATION

THERMODYNAMICS CYCLE

• Brayton Cycle (Power cycle)

THEORY & LAW INVOLVED

• **First Law of Thermodynamics**
Thermal energy is converted into kinetic energy which provides forwards thrust for airplane.

• **Second Law of Thermodynamics**
Thermal energy cannot be fully converted into kinetic energy, no jet engine can have 100 percent thermal efficiency.

• **Isobaric process**
Heat is absorbed by air to do work on the system, creating enough energy to turn blades.

• **Isentropic process**
Gas is expanded without any heat exchange, causing the blades to turn.

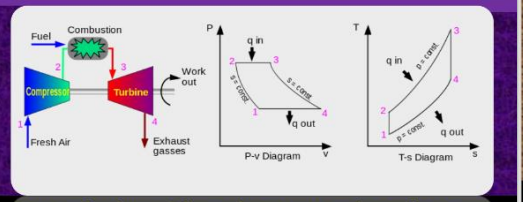
WORKING PRINCIPLE

• **Isentropic compression**
1) Ambient air is compressed by compressor.
2) The pressure of air increases.

• **Isobaric expansion**
1) Heat is added to compressed air through a fuel in combustion chamber.
2) The air is both heated and pressurized.

• **Isentropic expansion**
1) Heated and pressurized air turns the blades of turbine
2) Heated and pressurized air turns the blades of turbine

• **Isobaric compression**
1) Heat is released to surrounding
2) The temperature of turbine decreases.



Jet engine applies laws of thermodynamics to make airplane moves.

POWER INDUSTRY (THERMAL POWER PLANT)

NINTH APPLICATION

THERMODYNAMICS CYCLE

• Rankine Cycle (Power cycle)

THEORY & LAW INVOLVED

• **First Law of Thermodynamics**
Chemical energy is converted into thermal energy, then converted into mechanical energy, then converted into electrical energy.

• **Second Law of Thermodynamics**
For non-ideal Rankine cycle, the processes are irreversible because the entropy increases over time.

• **Isobaric process**
Heat is absorbed and released by water to change from liquid to steam and steam to liquid.

• **Isentropic process**
Steam is expanded to do work which causes the blades of turbine turning and compressed to operating pressure.

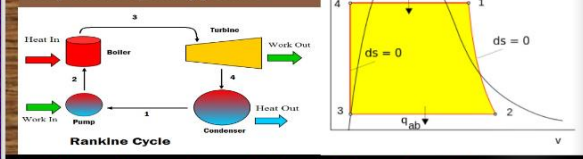
WORKING PRINCIPLE

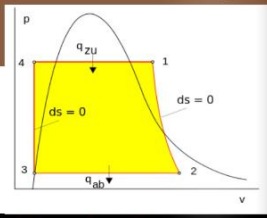
• **Isentropic compression**
1) Work is done to compress the working fluid by pump.
2) The pressure of water increases.

• **Isobaric expansion**
1) Heat is added to water through a fuel in boiler.
2) Water is converted into steam.

• **Isentropic expansion**
1) Steam expands and turns the blades of turbine.
2) Work is generated and converted into electrical energy.

• **Isobaric compression**
1) Heat is released from condenser to surrounding.
2) Steam is condensed into liquid.





Thermal power plant applies laws of thermodynamics to generate electricity.

RESEARCH INDUSTRY (MAGNETIC COOLING)

TENTH APPLICATION

THERMODYNAMICS CYCLE

• Similar to Carnot Cycle

THEORY & LAW INVOLVED

• **Third Law of Thermodynamics**
This method can achieve 0.3K but never 0K because of the constrain of this law.

• **Isomagnetic process**
To absorb or release heat from material or to surrounding.

• **Adiabatic process**
To increase and decrease the magnetic field. (Carnot cycle changes pressure instead of magnetic field)

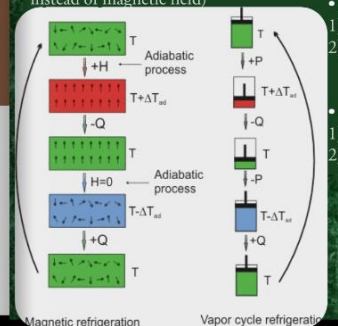
WORKING PRINCIPLE

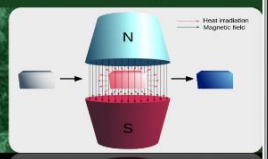
• **Adiabatic magnetization**
1) A magnetocaloric substance is placed in an insulated environment and external magnetic field is increased.
2) Magnetic dipoles of the atoms align and temperature of substance increases.

• **Isomagnetic enthalpic transfer**
1) Heat is removed by liquid helium and magnetic field is held constant
2) Magnetocaloric substance is cooled.

• **Adiabatic demagnetization**
1) Magnetic field is decreased.
2) Thermal energy causes the magnetic moments to overcome the field and substance cool down.

• **Isomagnetic entropic transfer**
1) Magnetic field is held constant.
2) Heat flows to working material.





Magnetic cooling applies laws of thermodynamics to attain extremely low temperature.

I. Overview and basic concepts



Otto von Guericke (1602 – 1686)

Thermodynamics as a scientific discipline

- 1650: **Otto von Guericke** (German scientist) built and designed the first vacuum pump
- 1656: **Robert Boyle** (Anglo-Irish physicist and chemist) in coordination with Robert Hooke (English scientist) built an air pump.
- 1679: **Denis Papin** (French physicist and mathematician) built a steam digester

I. Overview and basic concepts

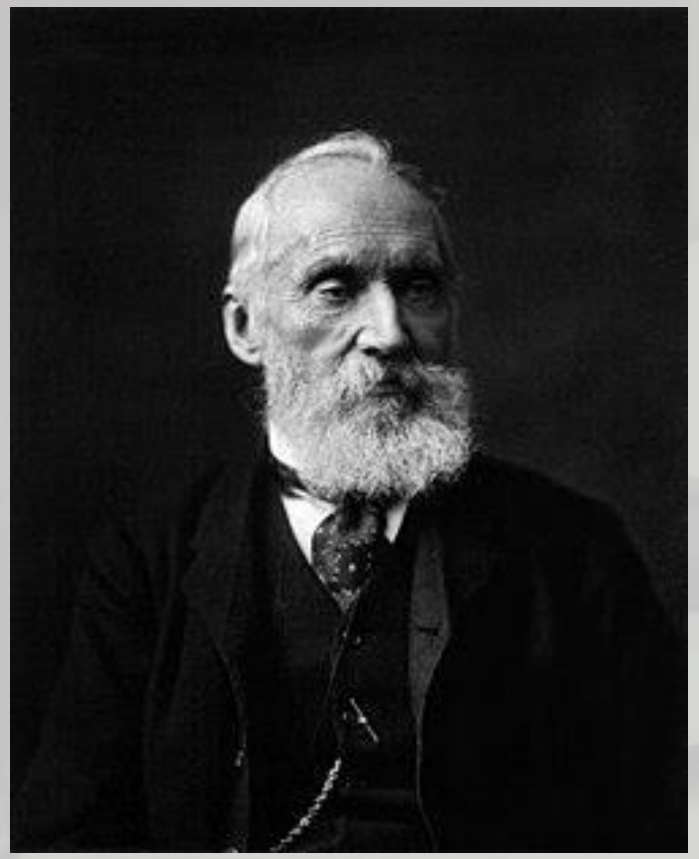


Nicolas Léonard Sadi Carnot
(1796 – 1832)

Thermodynamics as a scientific discipline

- 1697: **Thomas Savery** (English inventor and engineer) built the first engine
- 1824: **Sadi Carnot** (French engineer) - "father of thermodynamics", published *Reflections on the Motive Power of Fire*, a discourse on heat, power, energy and engine efficiency.

I. Overview and basic concepts

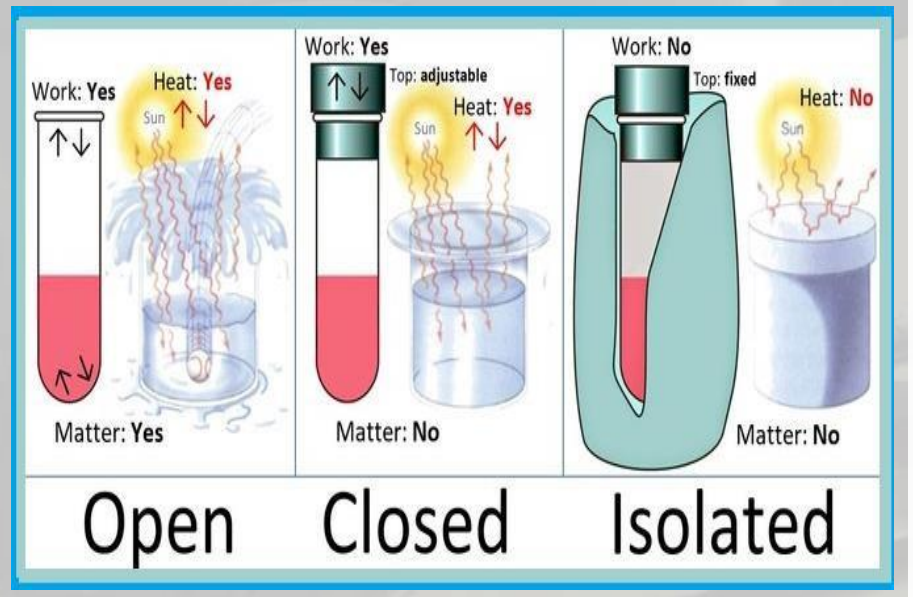


William Thomson
(1824 – 1907)

Thermodynamics as a scientific discipline

- **1873–76:** Josiah Willard Gibbs (American mathematical physicist) showed how thermodynamic processes could be graphically analyzed
- Early 20th century: **Gilbert N. Lewis, Merle Randall, and E. A. Guggenheim** (chemists) applied the mathematical methods of Gibbs to the analysis of chemical processes.

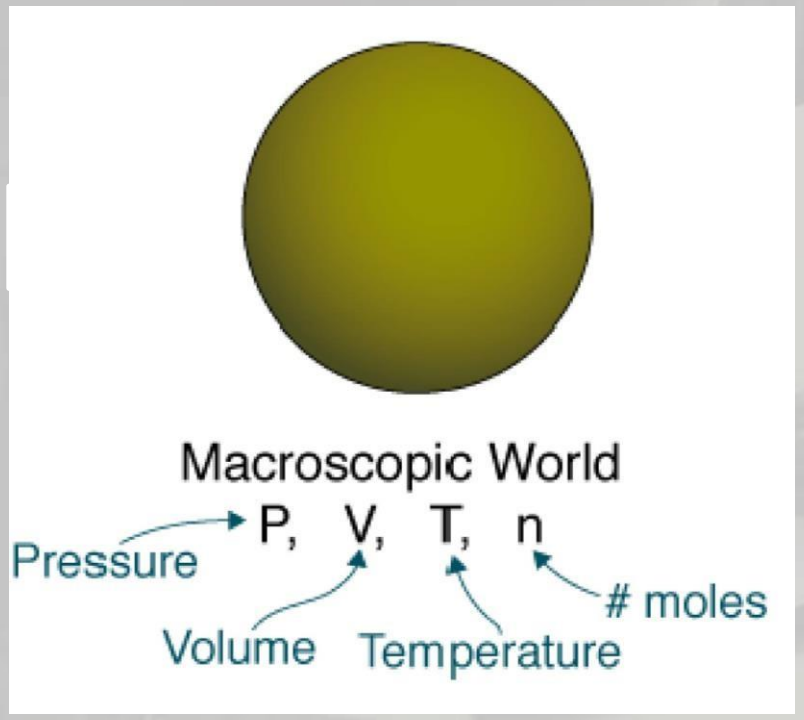
I. Overview and basic concepts



Basic concepts

- **Open system:** both energy interactions and mass interactions take place
- **Closed system:** only energy interactions and no mass interactions take place
- **Isolated system:** no energy interactions and no mass interactions take place

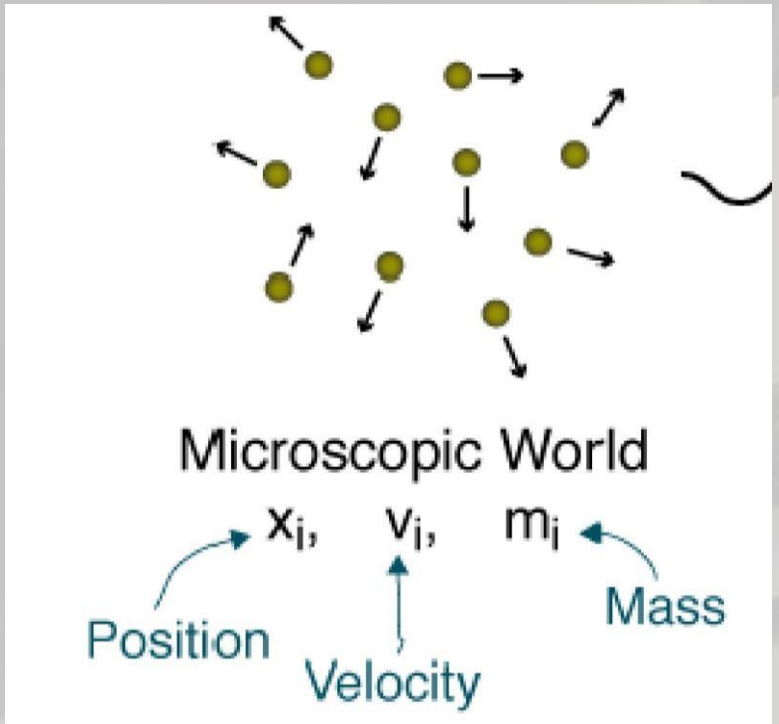
I. Overview and basic concepts



Basic concepts

- **Macroscopic system** is concerned with the effects of a certain number of matter, and these effects can be perceived by human senses.
- **Macroscopic properties**: The properties associated with a macroscopic system, i.e. pressure, volume, temperature, density, composition, viscosity, surface tension, refractive index, colour, etc.

I. Overview and basic concepts



Basic concepts

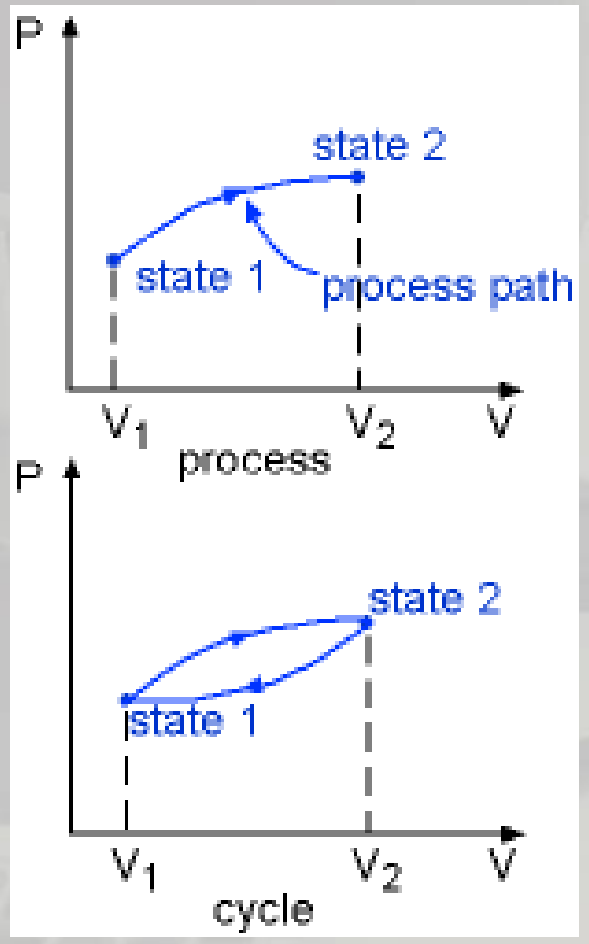
- **Microscopic system** is concerned with the effects of the action of many molecules, and these effects cannot be perceived by human senses.
- **Microscopic properties**: The properties associated with a microscopic system, i.e. molar mass, velocity of atoms or molecules, momentum, kinetic energy, etc.



Basic concepts

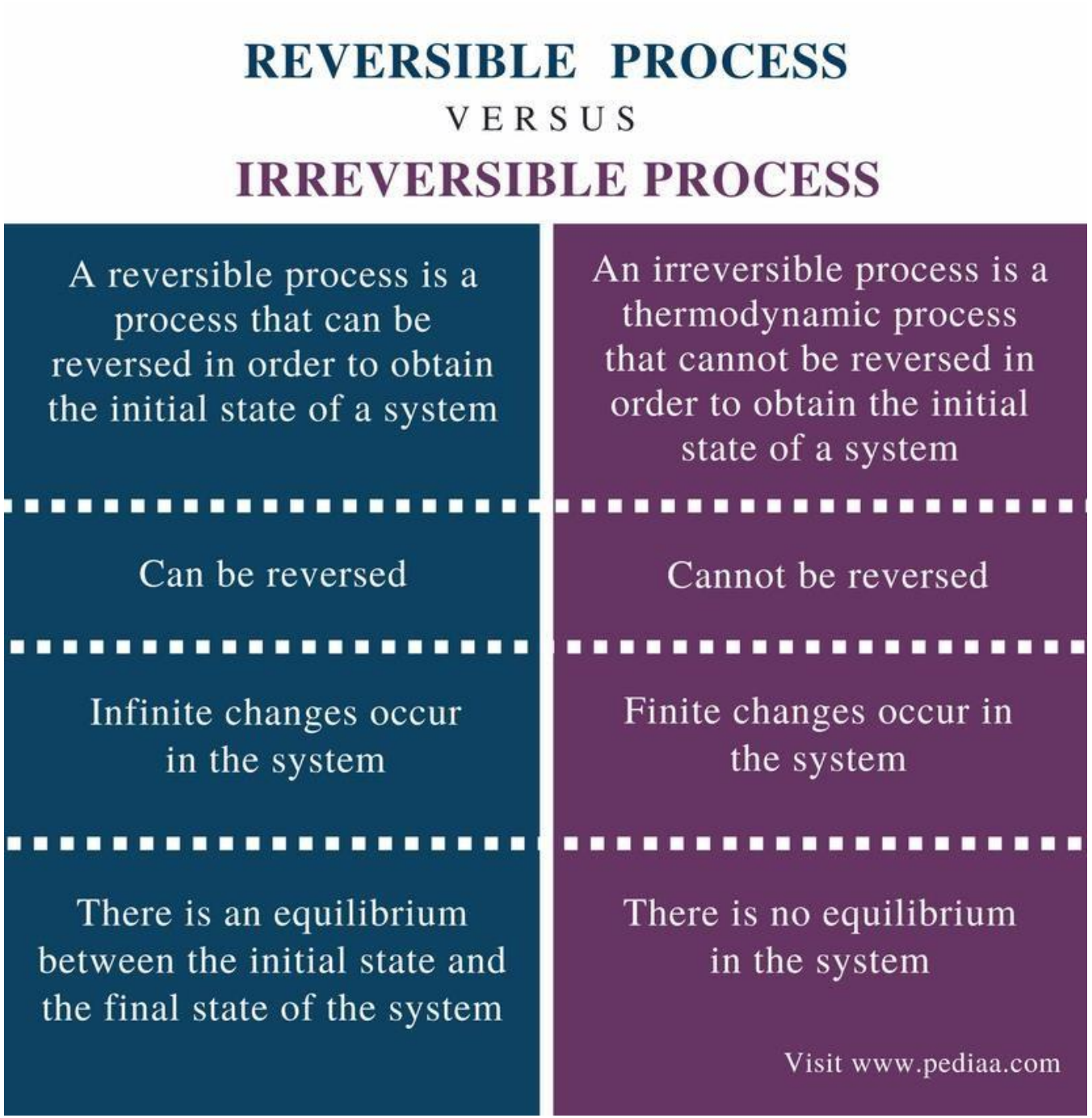
- **Intensive (intrinsic) properties:** properties that are independent of the mass of the system
- **Extensive (extrinsic) properties**: properties that are independent of the mass of the system

I. Overview and basic concepts



Basic concepts

- **State:** defined by specifying values of a set of measurable **properties** sufficient to determine all other properties. Ex: The state of fluid systems requires typical properties are (V, T, P).
- **Process:** any change of state of the system
- **Process path:** infinite states through which the system passes while going from initial state to final state



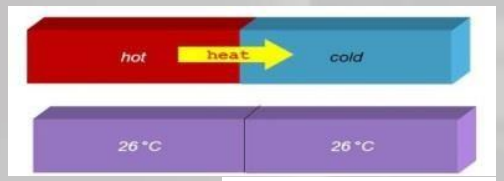
Basic concepts

- **Reversible process:** Ex: uniform and slow expansion or compression of a fluid in a piston

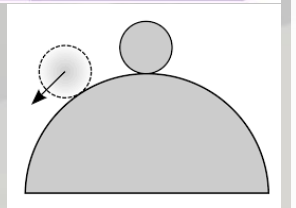
- **Irreversible process:** Ex: Heat transfer, Diffusion Electricity flow through a resistance

I. Overview and basic concepts

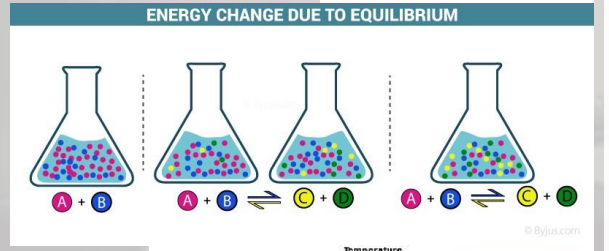
Thermal Equi.



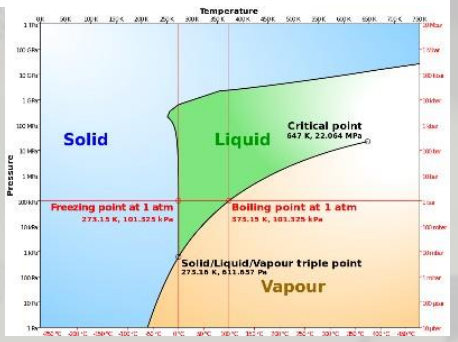
Mechanical Equi.



Chemical Equi.



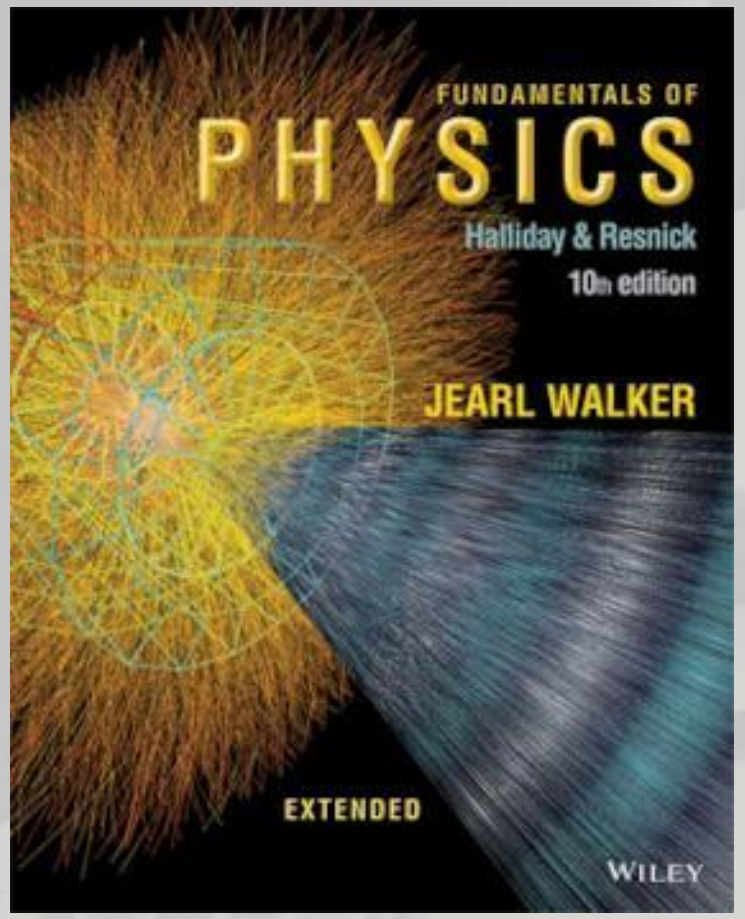
Phase Equi.



Basic concepts

- **Thermodynamic equilibrium:** there is no tendency for the state of a system to change spontaneously.
- Ex: the gas in a cylinder with a movable piston will be at equilibrium if T & P inside are uniform and if the restraining force on the piston is just sufficient to keep it from moving.

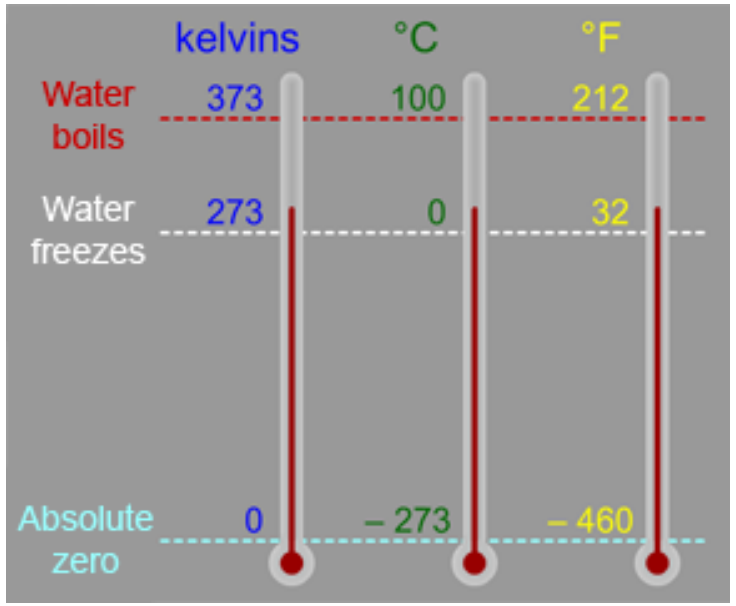
II. Temperature, heat and The First law of thermodynamics

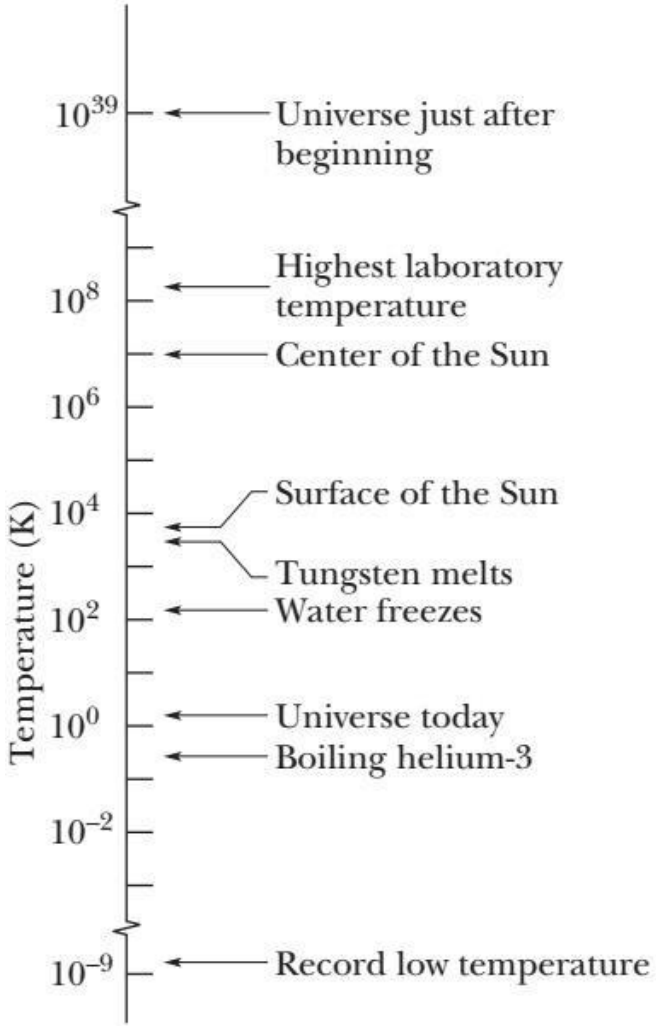


Outcomes

- Understand the concept of temperature
- Understand the concept of heat
- Apply the first law of thermodynamics
- Understand heat transfer mechanisms

Temperature





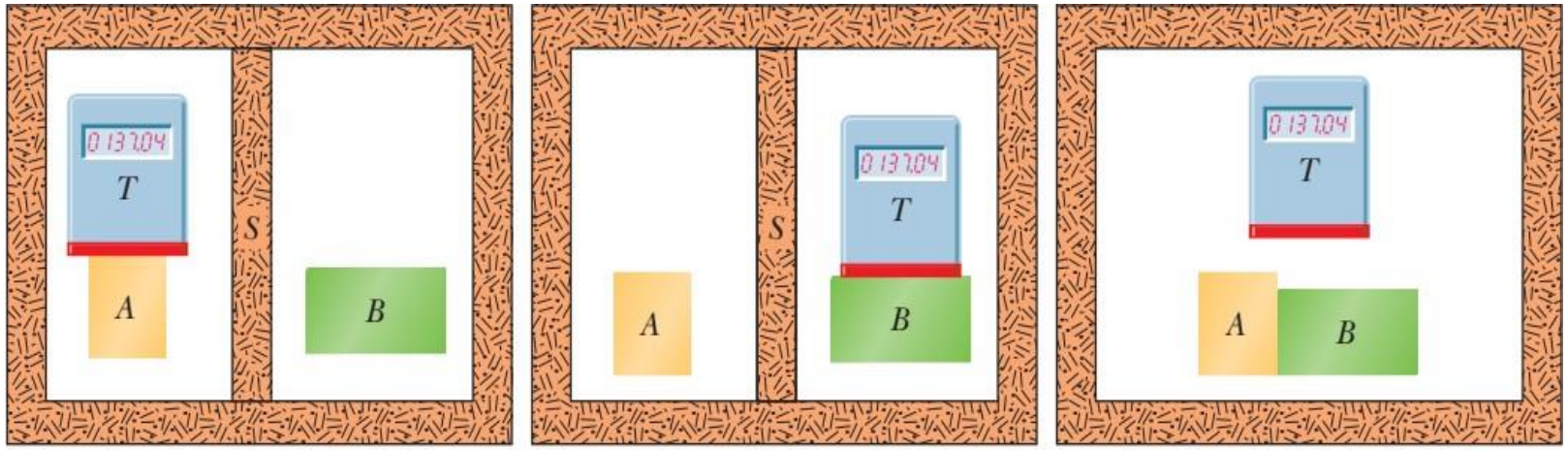
Temperature

- The degree or intensity of heat present in a substance or object
- Express the "hotness" or "coldness" of matter or radiation.
- 1 of the 7 SI quantities
- Conversions of temperature scales:

$$T_K = T_C + 273.15$$

$$T_C = (5/9)(T_F - 32)$$

Temperature



(a) (b) (c)

Zeroth law

- "If bodies A and B are each in thermal equilibrium with a third body T, then A and B are in thermal equilibrium with each other"
- In less formal language: When two bodies are in thermal equilibrium, their temperatures are equal.
- Came to light in the 1930s, long after the 1st and 2nd laws had been discovered and numbered.

MICROWAVE OVEN

THIRD APPLICATION



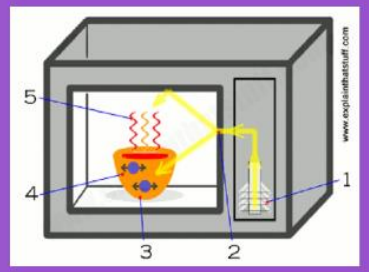
THEORY & LAW INVOLVED

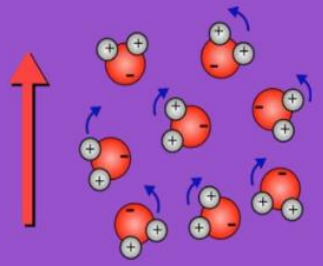
- **Zeroth Law of Thermodynamics**
  Microwave oven spins food slowly to cook food evenly and thermal equilibrium can be reached.
- **First Law of Thermodynamics**
  Radiant energy is converted into thermal energy which is used to cook food.
- **Conduction**
  Heat energy is transmitted through collisions between water molecules.
- **Convection**
  A fan is employed to circulate the hot air to cook the food.
- **Radiation**
  Microwave radiation is emitted to agitate the molecules of food which produces heat energy without physical contact.

WORKING PRINCIPLE

- **How microwave oven cooks food?**

1) Magnetron generates microwave by using electricity.
2) Magnetron blasts these waves into the food compartment through a channel called a wave guide.
3) Turntable is spinning slowly round so the microwaves can cook the food evenly.
4) Microwaves bounce back and forth off, penetrating the molecules to vibrate them.
5) Vibration of molecules produces heat energy to cook the food.

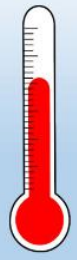




Microwave oven applies laws of thermodynamics to heat and cook the food

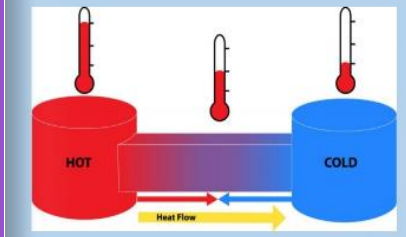
THERMOMETER

FIRST APPLICATION



THEORY & LAW INVOLVED

- **Zeroth Law of Thermodynamics**
  Temperature of thermometer indicates the temperature of certain object when thermometer reaches thermal equilibrium with that object.
- **First Law of Thermodynamics**
  Heat energy is converted into kinetic energy which will cause expansion of mercury.
- **Second Law of Thermodynamics**
  Heat will flow from thermometer to object or object to thermometer depending on their temperature.
- **Thermal expansion**
  Mercury will expand when heated because the molecules move faster and take more space.



WORKING PRINCIPLE

- **When temperature of thermometer is greater than temperature of object**

1) When thermometer and object in contact with each other, heat energy flows from object to thermometer.
2) Heat energy is converted into kinetic energy gained by mercury molecules in thermometer, causing expansion of thermometer.
3) This process continues until thermal equilibrium is reached when temperature of thermometer is same as temperature of object.

- **When temperature of thermometer is lower than temperature of object**

1) When thermometer and object in contact with each other, heat energy flows from thermometer to object.
2) Heat energy is lost as kinetic energy by mercury molecules in thermometer, causing contraction.
3) This process continues until thermal equilibrium is reached when temperature of thermometer is same as temperature of object.

Thermometer applies laws of thermodynamics to measure the temperature of object.

Zeroth law

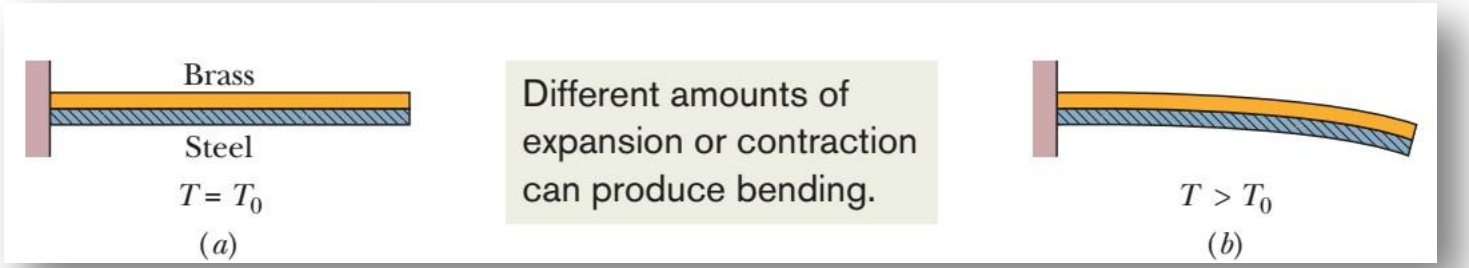
- "If bodies A and B are each in thermal equilibrium with a third body T, then A and B are in thermal equilibrium with each other"

Temperature

Linear expansion: If temperature of a metal rod of length L is raised by an amount ΔT, its length is found to increase by:

$$\Delta L = L\alpha\,\Delta T$$

α: coefficient of linear expansion



Volume expansion: If the temperature of a solid or liquid whose volume is V is increased by an amount ΔT, the increase in volume is:

$$\Delta V = V\beta\,\Delta T$$

β: coefficient of volume expansion

Thermal expansion

- The tendency of matter to change its shape, area, volume and density in response to a change in temperature
- With the added energy, the atoms can move a bit farther from one another than usual, against the spring-like interatomic forces that hold every solid together

Temperature



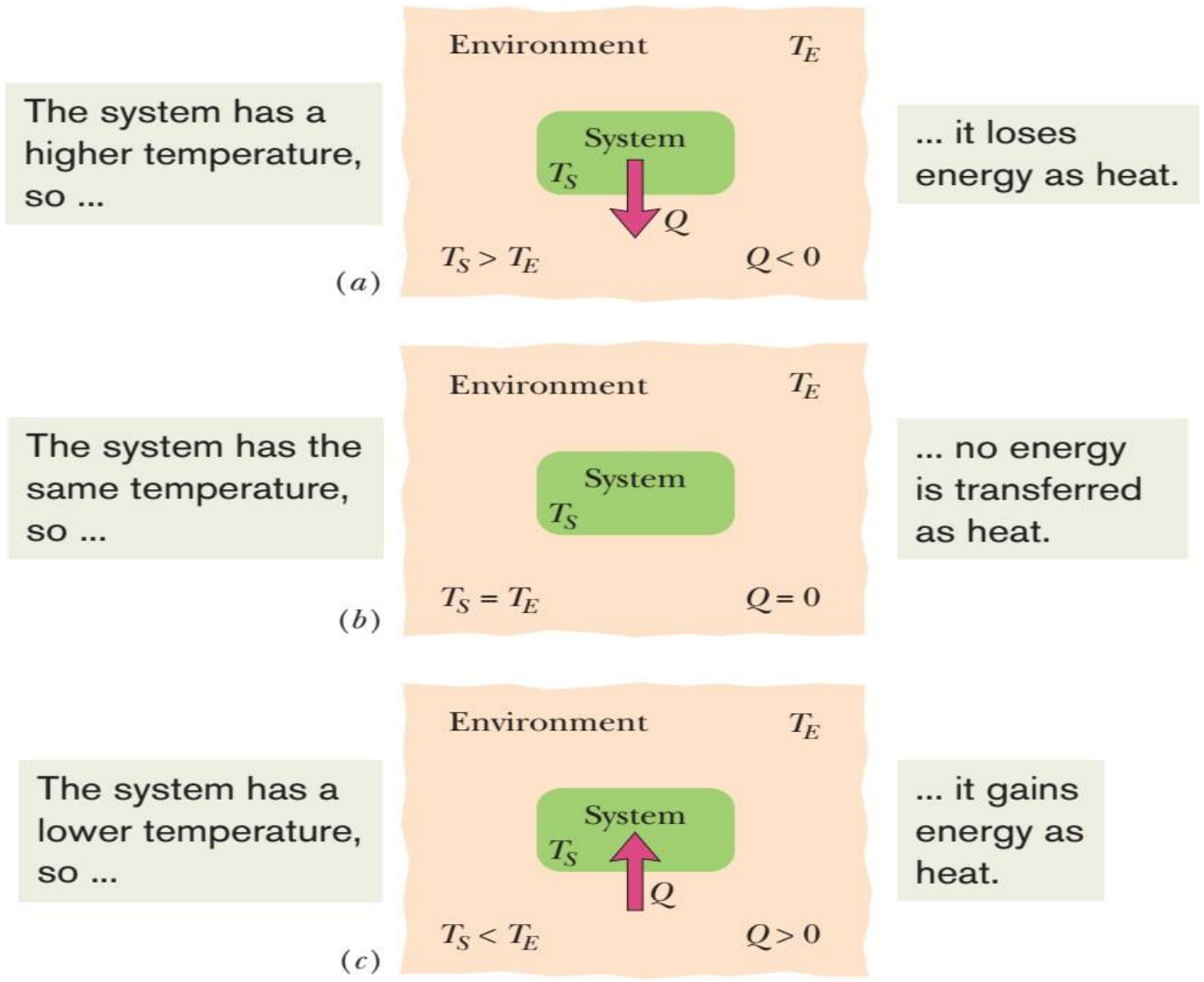
Truck buckling due to thermal expansion



Thermal expansion

- **Thermal expansion** of materials with an increase in temperature must be anticipated in many common situations.

Heat



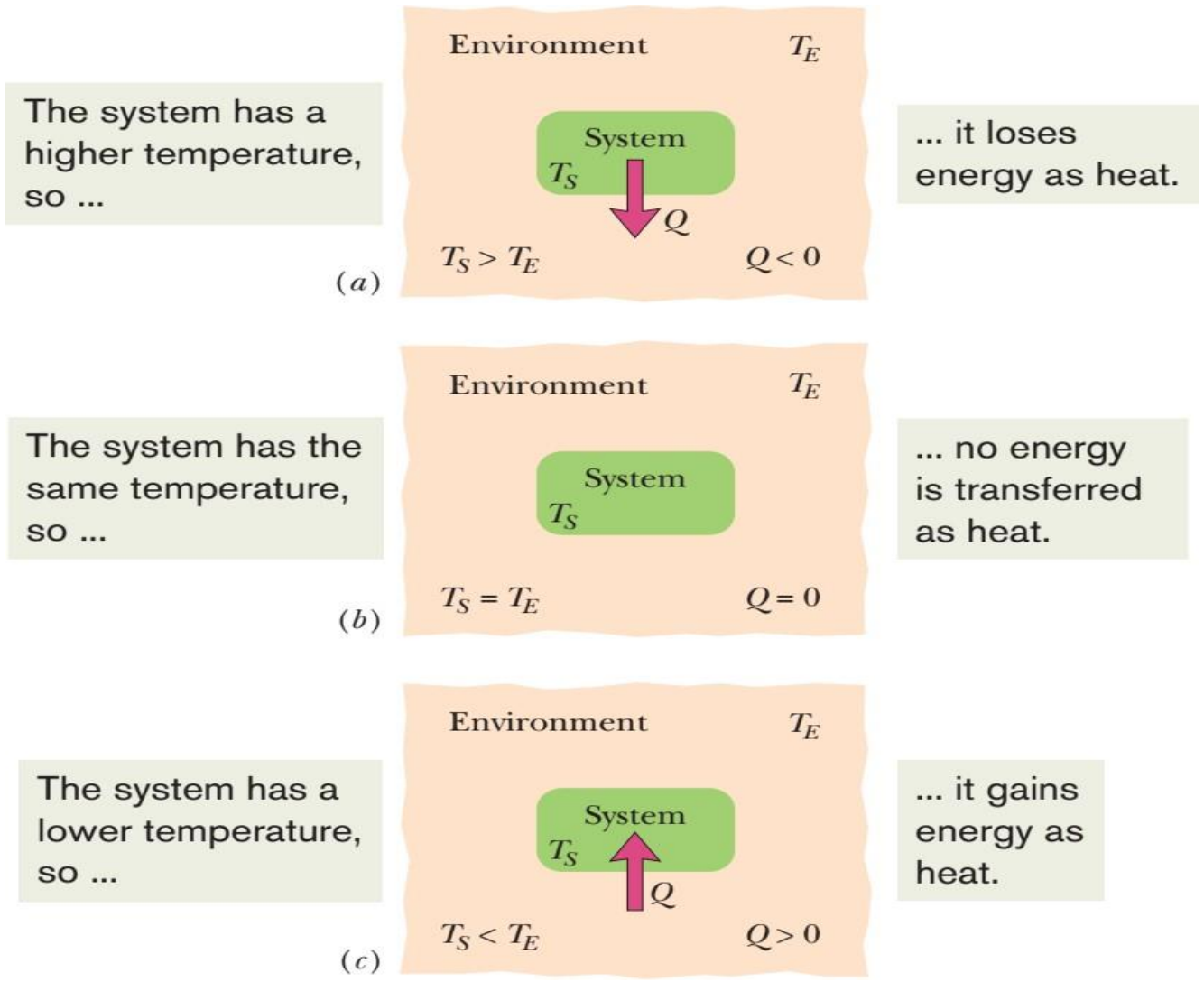
Heat

The energy transferred from one medium or object to another because of a temperature difference that exists between them

Is heat an intrinsic property of a system?

No! They have meaning only as they describe the transfer of energy into or out of a system.

Heat



Heat

Unit of heat?

Before 1948: **Calorie**/Btu (Before scientists realized that heat is transferred energy, heat was measured in terms of its ability to raise temperature of water)

After 1948: **Joule** (Scientists decided that heat (like work) is transferred energy)

$1\text{cal} = 3.968\text{x } 10^{-3}\ \text{Btu}$
$= 4.1868\ \text{J}$

Specific Heat

If heat Q is absorbed by an object, the object's temperature change Tf and Ti is related to Q by:

$$Q = C\,\Delta T = C(T_f - T_i),$$

If the object has mass m :

$$Q = cm\,\Delta T = cm(T_f - T_i).$$

- ***C* : heat capacity** of an object (J/K)
- ***c* : specific heat** of a unit mass of material of which the object is made (J/kg.K)

Specific Heat

IMPORTANT: When using the specific heat of any substance, we need to know the conditions under which energy is transferred as heat.

For solids and liquids, we usually assume that the sample is:

- Under constant pressure (usually atmospheric) during the transfer.
- Held at constant volume while the heat is absorbed.

→Cv and Cp differ usually by no more than a few percent.

For gases: quite different values for their specific heats under constant-pressure conditions and under constant-volume conditions.

Substance	Specific Heat cal / g·K	Specific Heat J / kg·K	Molar Specific Heat J / mol·K
Elemental Solids			
Lead	0.0305	128	26.5
Tungsten	0.0321	134	24.8
Silver	0.0564	236	25.5
Copper	0.0923	386	24.5
Aluminum	0.215	900	24.4
Other Solids			
Brass	0.092	380	
Granite	0.19	790	
Glass	0.20	840	
Ice (−10°C)	0.530	2220	
Liquids			
Mercury	0.033	140	
Ethyl alcohol	0.58	2430	
Seawater	0.93	3900	
Water	1.00	4187	

Heat of transformation

Melting. To *melt* a solid means to change it from the solid state to the liquid state

Vaporizing. To *vaporize* a liquid means to change it from the liquid state to the vapor (gas) state.

Amount of energy required to change the state - but not the temperature - of a particular material of mass *m*:

$$Q = Lm.$$

- ***L :*** *heat of transformation (J/Kg)*

- ***Lv :*** *Heat of vaporization*

Water at its normal boiling or condensation temperature:
L_v = 539 cal/g = 40.7 kJ/mol = 2256 kJ/kg

- ***L_f :*** *Heat of fusion*

Water at its normal freezing or melting temperature:
L_f = 79.5 cal/g = 6.01 kJ/mol = 333 kJ/kg



Exercise

(a) How much heat must be absorbed by ice of mass $m = 720$ g at -10°C to take it to the liquid state at 15°C?

(b) If we supply the ice with a total energy of only 210 kJ (as heat), what are the final state and temperature of the water?

Copyrights @ Hello Techie Vietnam



Copyrights @ Hello Techie Vietna

(a) How much heat must be absorbed by ice of mass m = 720 g at −10°C to take it to the liquid state at 15°C?

KEY IDEAS

The heating process is accomplished in three steps: (1) The ice cannot melt at a temperature below the freezing point—so initially, any energy transferred to the ice as heat can only increase the temperature of the ice, until 0°C is reached. (2) The temperature then cannot increase until all the ice melts—so any energy transferred to the ice as heat now can only change ice to liquid water, until all the ice melts. (3) Now the energy transferred to the liquid water as heat can only increase the temperature of the liquid water.

Warming the ice: The heat Q_1 needed to take the ice from the initial $T_i = -10°C$ to the final $T_f = 0°C$ (so that the ice can then melt) is given by Eq. 18-14 ($Q = cm\,\Delta T$). Using the specific heat of ice c_{ice} in Table 18-3 gives us

$$\begin{aligned} Q_1 &= c_{ice} m(T_f - T_i) \\ &= (2220\ \text{J/kg}\cdot\text{K})(0.720\ \text{kg})[0°C - (-10°C)] \\ &= 15\,984\ \text{J} \approx 15.98\ \text{kJ}. \end{aligned}$$

Melting the ice: The heat Q_2 needed to melt all the ice is given by Eq. 18-16 ($Q = Lm$). Here L is the heat of fusion L_F, with the value given in Eq. 18-18 and Table 18-4. We find

$$Q_2 = L_F m = (333\ \text{kJ/kg})(0.720\ \text{kg}) \approx 239.8\ \text{kJ}.$$

Warming the liquid: The heat Q_3 needed to increase the temperature of the water from the initial value $T_i = 0°C$ to the final value $T_f = 15°C$ is given by Eq. 18-14 (with the specific heat of liquid water c_{liq}):

$$\begin{aligned} Q_3 &= c_{liq} m(T_f - T_i) \\ &= (4186.8\ \text{J/kg}\cdot\text{K})(0.720\ \text{kg})(15°C - 0°C) \\ &= 45\,217\ \text{J} \approx 45.22\ \text{kJ}. \end{aligned}$$

Total: The total required heat Q_{tot} is the sum of the amounts required in the three steps:

$$\begin{aligned} Q_{tot} &= Q_1 + Q_2 + Q_3 \\ &= 15.98\ \text{kJ} + 239.8\ \text{kJ} + 45.22\ \text{kJ} \\ &\approx 300\ \text{kJ}. \end{aligned} \qquad \text{(Answer)}$$

Note that most of the energy goes into melting the ice rather than raising the temperature.

(b) If we supply the ice with a total energy of only 210 kJ (as heat), what are the final state and temperature of the water?

KEY IDEA

From step 1, we know that 15.98 kJ is needed to raise the temperature of the ice to the melting point. The remaining heat Q_{rem} is then 210 kJ − 15.98 kJ, or about 194 kJ. From step 2, we can see that this amount of heat is insufficient to melt all the ice. Because the melting of the ice is incomplete, we must end up with a mixture of ice and liquid; the temperature of the mixture must be the freezing point, 0°C.

Calculations: We can find the mass m of ice that is melted by the available energy Q_{rem} by using Eq. 18-16 with L_F:

$$m = \frac{Q_{rem}}{L_F} = \frac{194\ \text{kJ}}{333\ \text{kJ/kg}} = 0.583\ \text{kg} \approx 580\ \text{g}.$$

Thus, the mass of the ice that remains is 720 g − 580 g, or 140 g, and we have

580 g water and 140 g ice, at 0°C. (Answer)

The First law of Thermodynamics

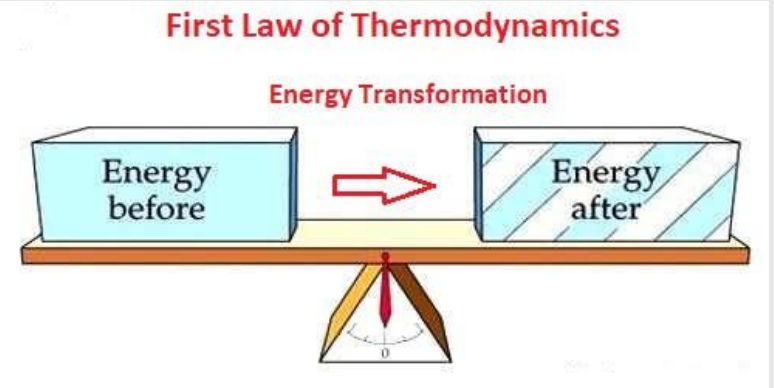
- A formulation of the law of conservation of energy: *the total energy of any isolated system, which cannot exchange energy or matter, is constant. Energy can be transformed from one form to another, but can be neither created nor destroyed.*

- Adapted for thermodynamic processes.

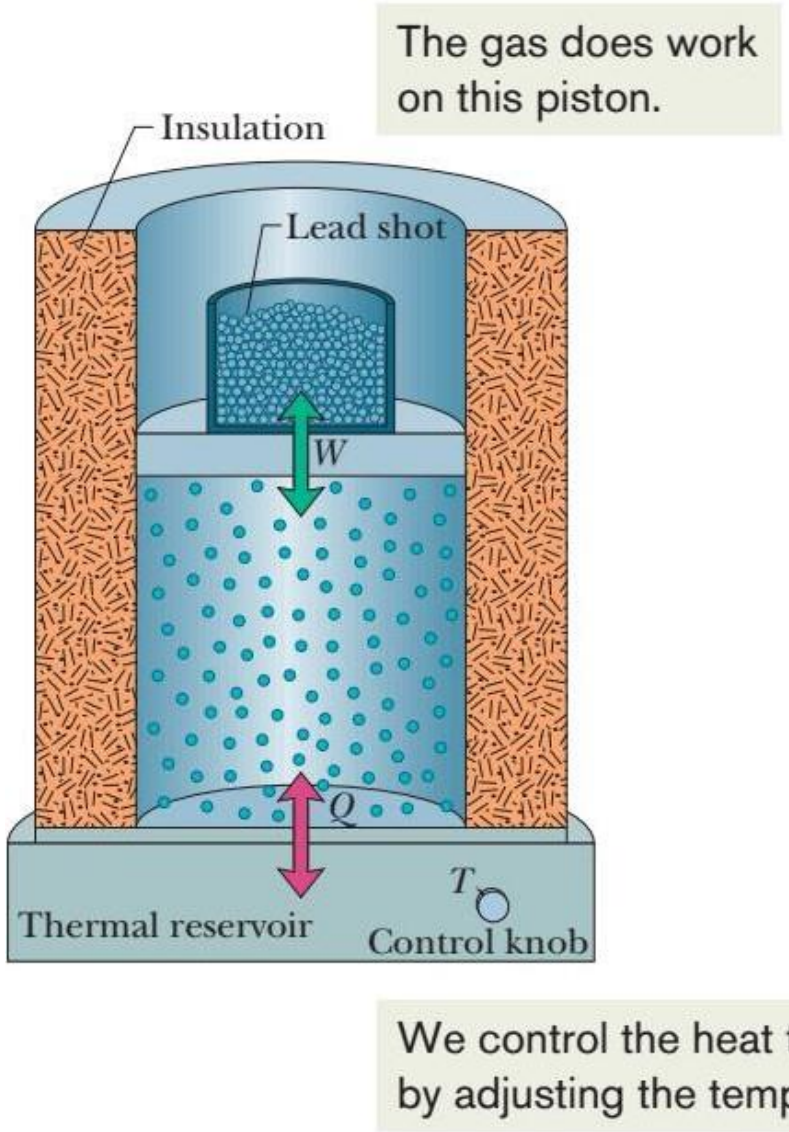
- It distinguishes two forms of energy transfer: **heat** and **thermodynamic work** for a system of a constant amount of matter.

- It defines the **internal energy** of a system, an extensive property for taking account of the balance of energies in the system.

How energy can be transferred as heat and work between a system and its environment?



The First law of Thermodynamics



Let consider an isolated system: A gas confined to a cylinder with a movable piston

The upward force on the piston = the weight of lead shot

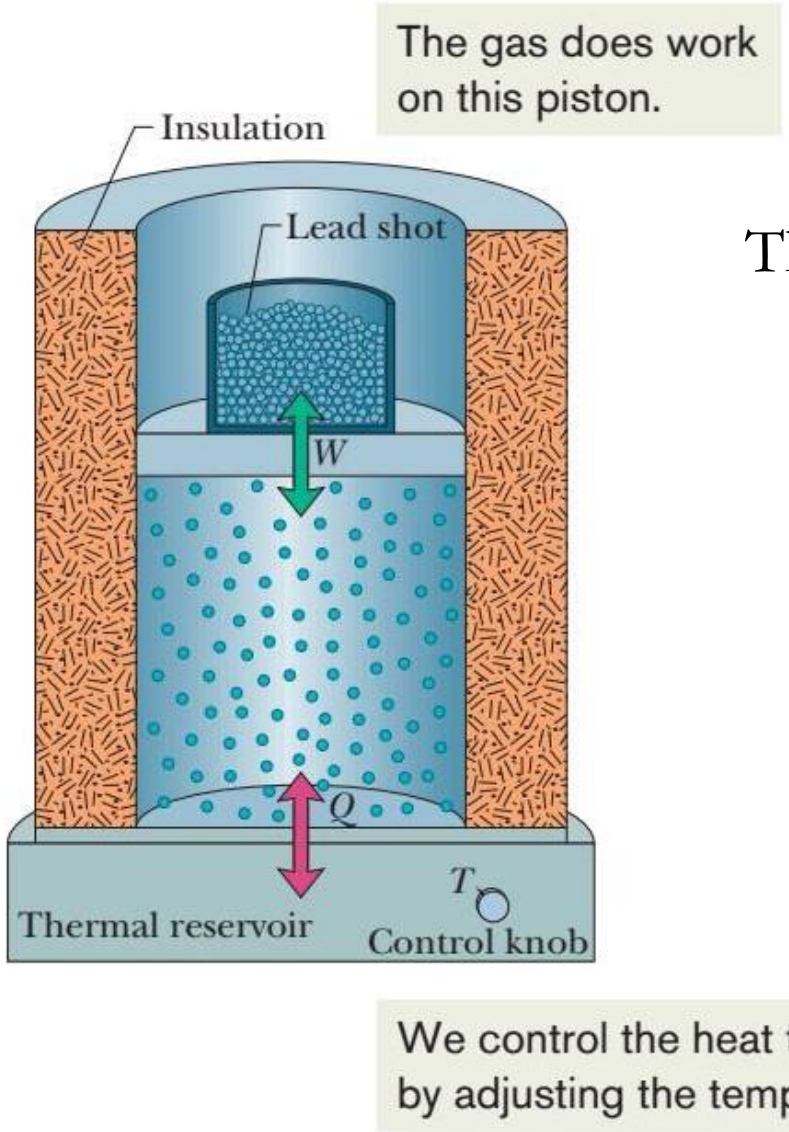
The bottom rests on a reservoir for thermal energy

The walls does not allow any transfer of energy as heat

Question:

How energy can be transferred as heat and work between a system and its environment?

The First law of Thermodynamics



The system (the gas) starts from:

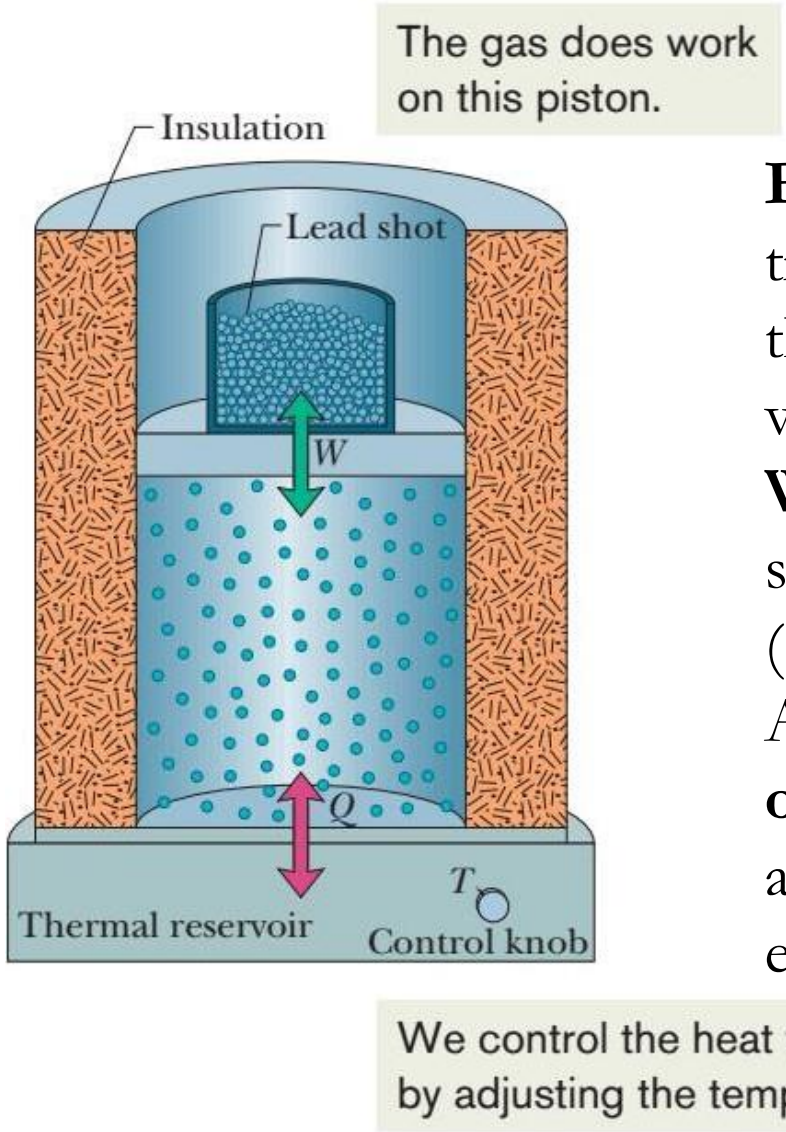
an **initial state** (P_i, V_i, T_i)
to
a **final state** (P_f, V_f, T_f)

Here, we perform
*a **thermodynamic process***

Question:

How energy can be transferred as heat and work between a system and its environment?

The First law of Thermodynamics



Energy as heat (Q) may be transferred into the system from the thermal reservoir (+ Q) or vice versa (- Q).

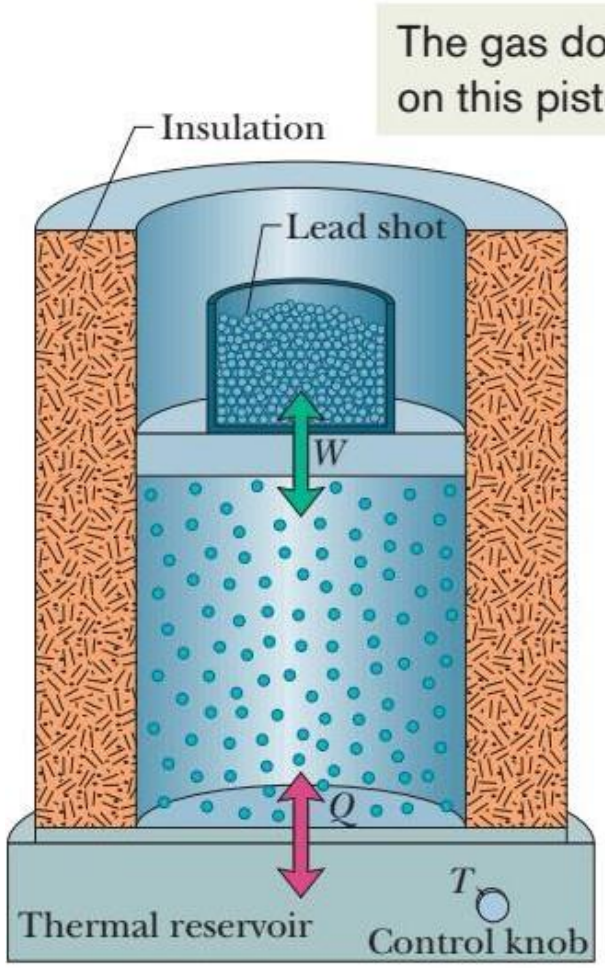
Work (W) can be done by the system to raise the loaded piston (+ W) or lower it (- W).

Assume that all such changes **occur slowly** -> the system is always in (approximate) thermal equilibrium

Question:

How energy can be transferred as heat and work between a system and its environment?

The First law of Thermodynamics



If a few lead shot is removed, the gas pushes the piston upward through a differential displacement $d\vec{s}$ with an upward force $\vec{F}$.

We can assume that $\vec{F}$ is constant during this tiny displacement. Then:

$$\vec{F} = pA$$

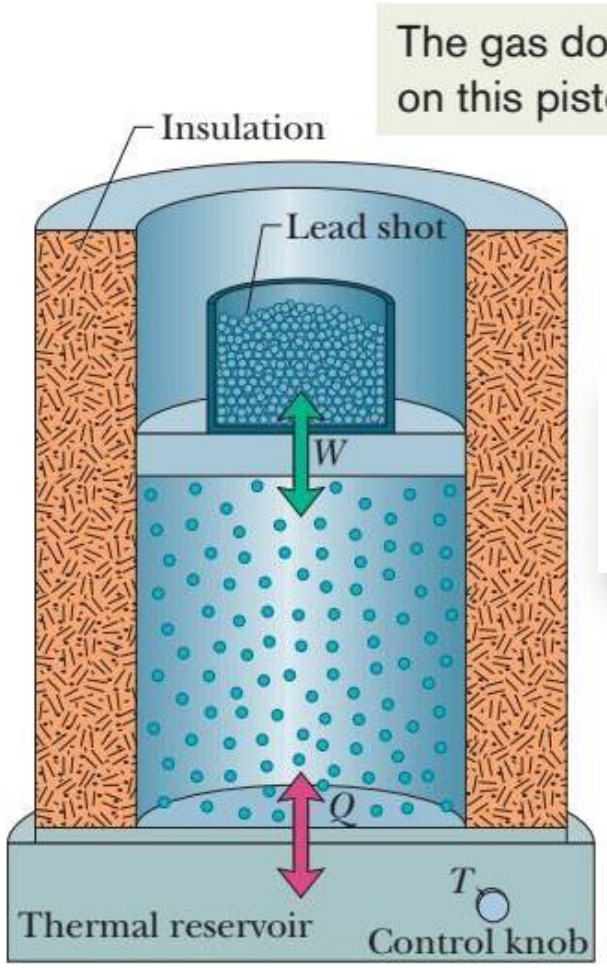
p: pressure of the gas

A: the face area of the piston.

Question:

How energy can be transferred as heat and work between a system and its environment?

The First law of Thermodynamics



Differential work dW done by the gas during displacement:

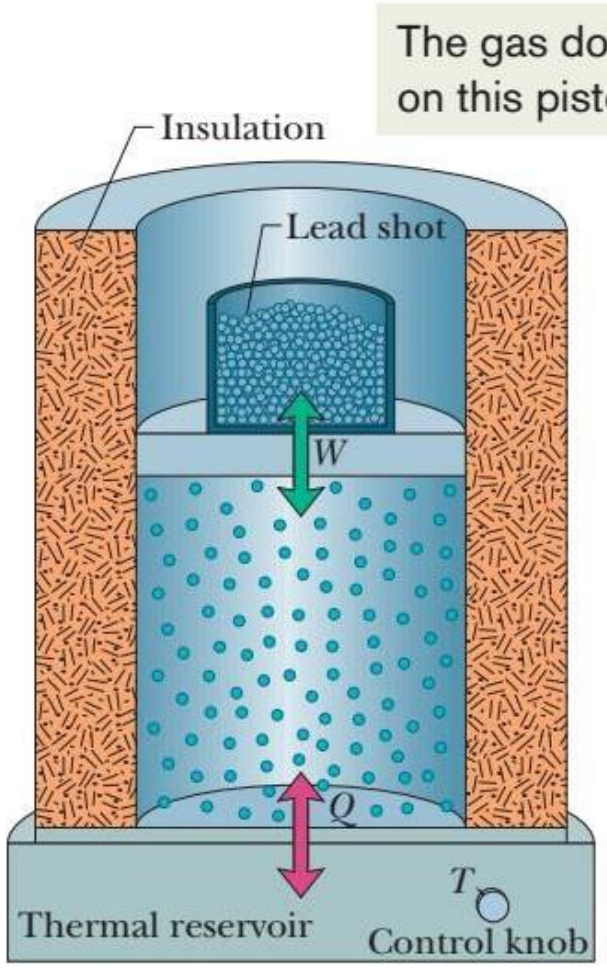
$$dW = \vec{F} \cdot d\vec{s} = (pA)(ds) = p(A\,ds)$$
$$= p\,dV$$

dV: differential change in gas volume due to piston's movement

Question:

How energy can be transferred as heat and work between a system and its environment?

The First law of Thermodynamics



When enough shot is removed to allow the gas to change from V_i to V_f:

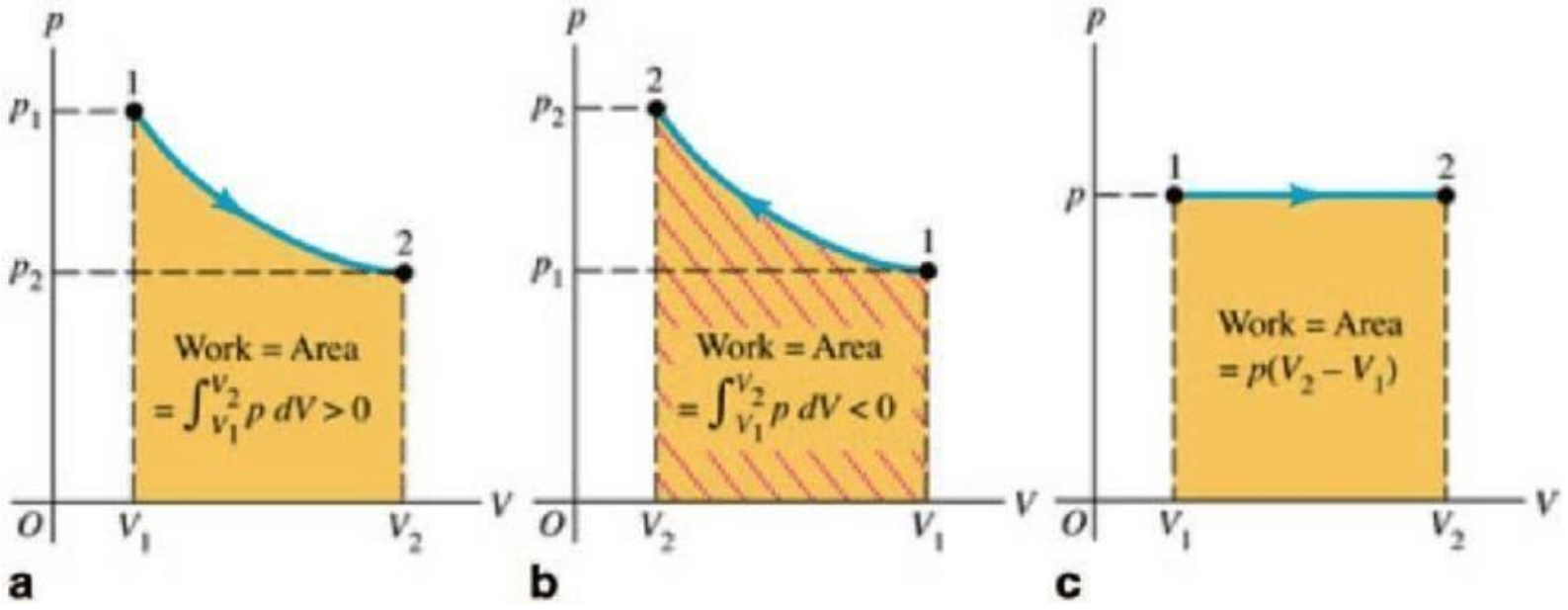
$$W = \int dW = \int_{V_i}^{V_f} p\, dV.$$

To evaluate the work, we need to know how pressure varies with volume for the actual process by which the system changes from state i to state f

Question:

How energy can be transferred as heat and work between a system and its environment?

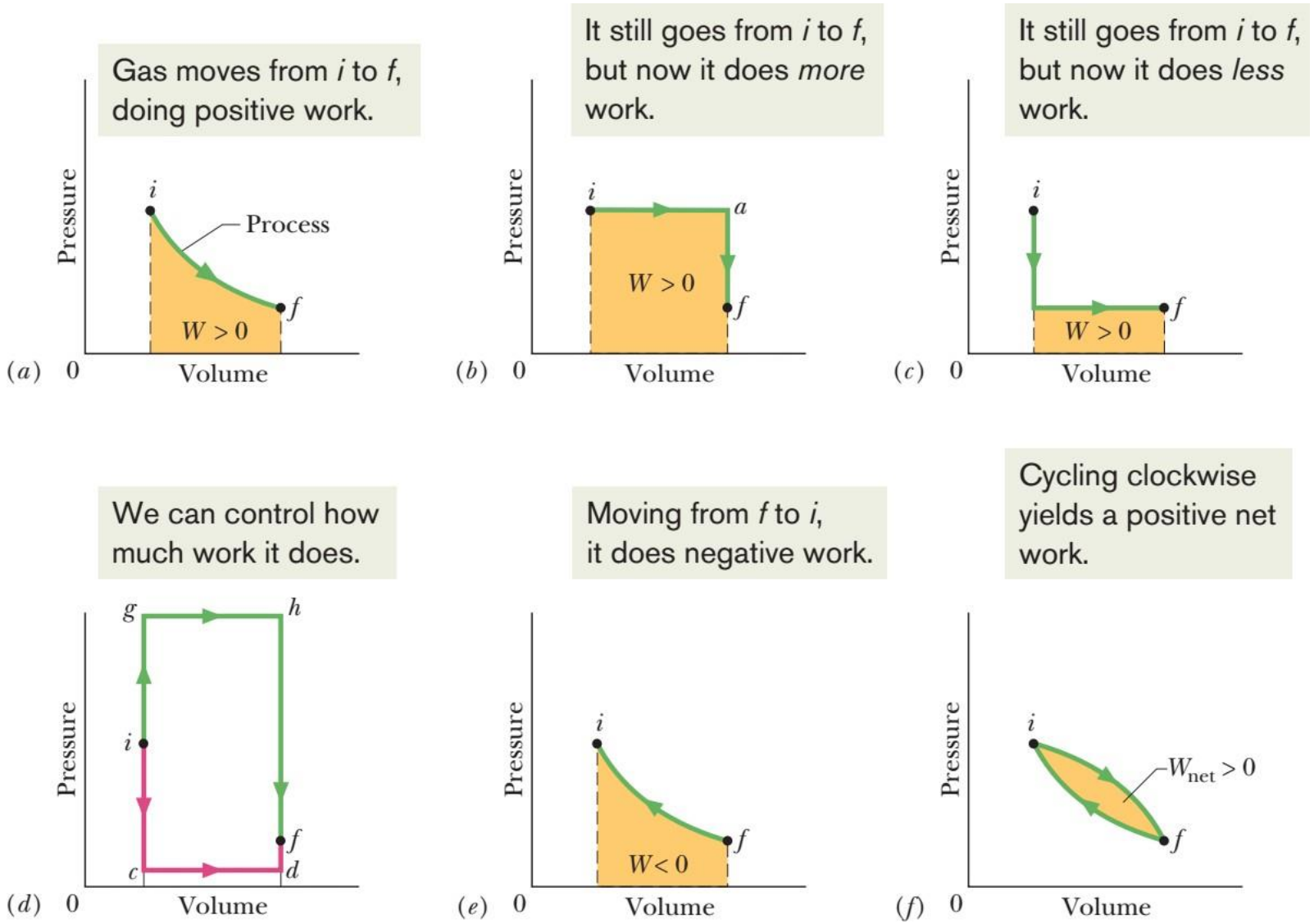
The First law of Thermodynamics



PV diagram:

A pressure–volume diagram (or PV diagram, or volume–pressure loop) is **used to describe corresponding changes in volume and pressure in a system**

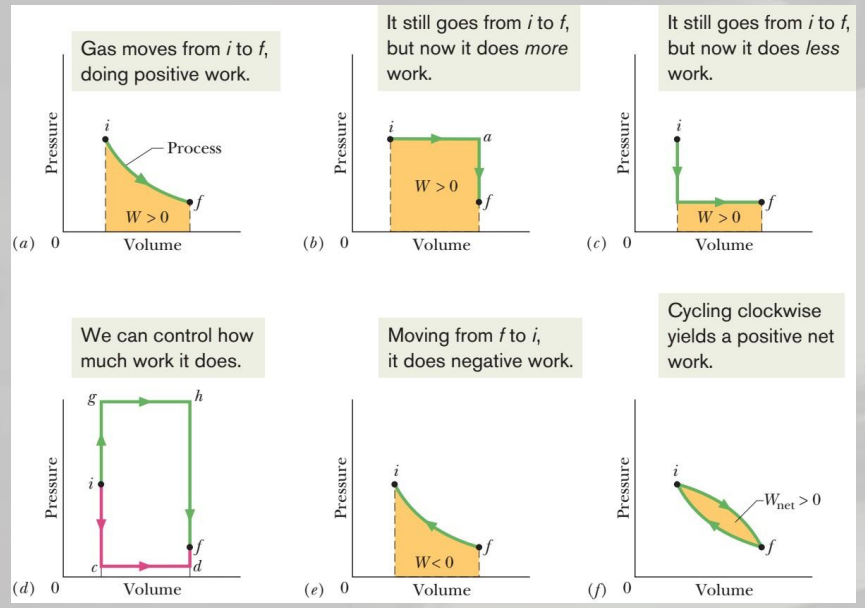
The First law of Thermodynamics



PV diagram:

A pressure–volume diagram (or PV diagram, or volume–pressure loop) is **used to describe corresponding changes in volume and pressure in a system**

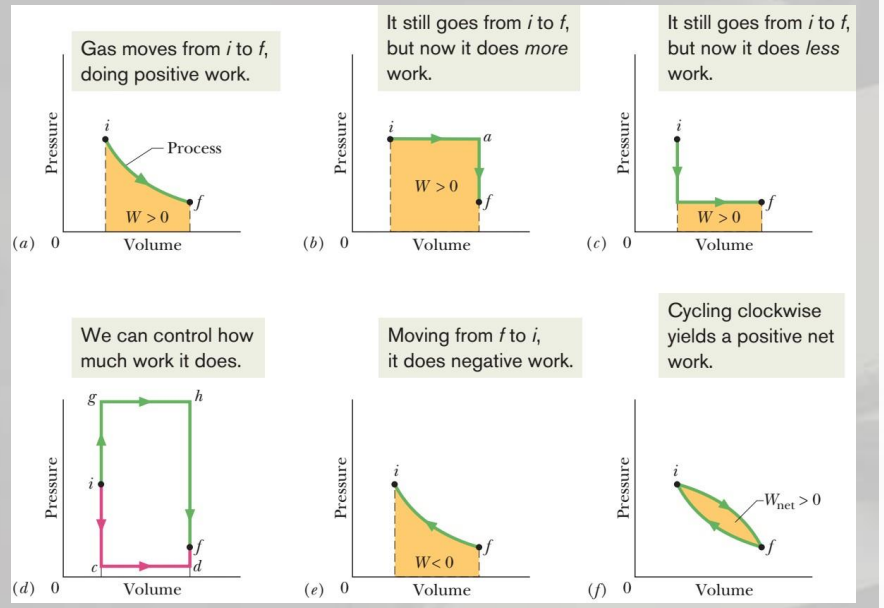
The First law of Thermodynamics



To sump up:

- A system can be taken from a given initial state to a given final state by an infinite number of processes.
- Heat *Q* and work W are *path-dependent* quantities.

The First law of Thermodynamics



However…

- $Q - W$ is the same for all processes!
- Must represent a change in some intrinsic property of the system: **Internal energy** E_{int}

The First law of Thermodynamics

The first law of thermodynamics

$$\Delta E_{int} = E_{int,i} - E_{int,f} = Q - W$$

The internal energy $\mathbf{E_{int}}$ of a system tends to increase if energy is **added as heat Q** and tends to decrease if energy is lost as work **W done by the system**

The First law of Thermodynamics

The first law of thermodynamics

$$\Delta E_{int} = E_{int,i} - E_{int,f} = Q - W$$

BE CARREFUL OF THE SIGN!!

Q positive heat added to the system
Q negative: Heat lost from the system
W positive : work done by the system
W negative: work done on the system

The First law of Thermodynamics

AUTOMOTIVE INDUSTRY (HEAT ENGINE)

SEVENTH APPLICATION

THERMODYNAMICS CYCLE

• Carnot Cycle (Power cycle)

THEORY & LAW INVOLVED

• **First Law of Thermodynamics**
Heat energy can be converted into mechanical energy, no heat engine can have thermal efficiency greater than 100 percent.

• **Second Law of Thermodynamics**
Heat energy cannot be fully converted into mechanical energy, no heat engine can have 100 percent thermal efficiency.

• **Isothermal process**
Heat is absorbed and released by gas to do work on or by the system, causing piston moving up and down.

• **Adiabatic process**
Gas is expanded and compressed to do on or by the thermally insulated system, causing the piston moving up and down.

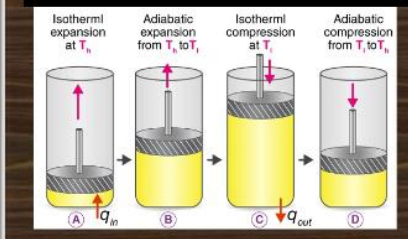
WORKING PRINCIPLE

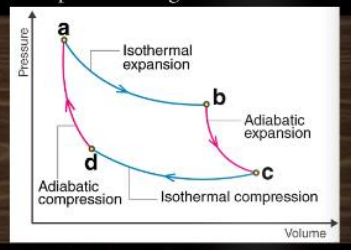
• **Isothermal expansion**
1) Gas absorbs heat at a constant temperature.
2) Gas does work on the system and expands.

• **Adiabatic expansion**
1) Gas continues expanding without heat transfer
2) Gas does work on the system and temperature of gas decreases.

• **Isothermal compression**
1) Gas releases heat a constant temperature
2) **Work done by system and gas is compressed**

• **Adiabatic compression**
1) Gas continues compressing without heat transfer.
2) Work done by system and temperature of gas increases.





Heat engine applies laws of thermodynamics to make car moves.

AEROSPACE INDUSTRY (JET ENGINE)

EIGHTH APPLICATION

THERMODYNAMICS CYCLE

• Brayton Cycle (Power cycle)

THEORY & LAW INVOLVED

• **First Law of Thermodynamics**
Thermal energy is converted into kinetic energy which provides forwards thrust for airplane.

• **Second Law of Thermodynamics**
Thermal energy cannot be fully converted into kinetic energy, no jet engine can have 100 percent thermal efficiency.

• **Isobaric process**
Heat is absorbed by air to do work on the system, creating enough energy to turn blades.

• **Isentropic process**
Gas is expanded without any heat exchange, causing the blades to turn.

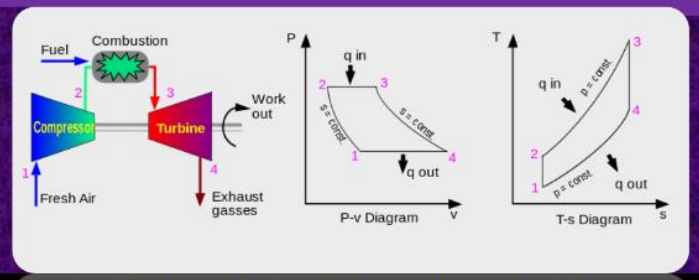
WORKING PRINCIPLE

• **Isentropic compression**
1) Ambient air is compressed by compressor.
2) The pressure of air increases.

• **Isobaric expansion**
1) Heat is added to compressed air through a fuel in combustion chamber.
2) The air is both heated and pressurized.

• **Isentropic expansion**
1) Heated and pressurized air turns the blades of turbine
2) Heated and pressurized air turns the blades of turbine

• **Isobaric compression**
1) Heat is released to surrounding
2) The temperature of turbine decreases.



Jet engine applies laws of thermodynamics to make airplane moves.

POWER INDUSTRY (THERMAL POWER PLANT)

NINTH APPLICATION

THERMODYNAMICS CYCLE

• Rankine Cycle (Power cycle)

THEORY & LAW INVOLVED

• **First Law of Thermodynamics**
Chemical energy is converted into thermal energy, then converted into mechanical energy, then converted into electrical energy.

• **Second Law of Thermodynamics**
For non-ideal Rankine cycle, the processes are irreversible because the entropy increases over time.

• **Isobaric process**
Heat is absorbed and released by water to change from liquid to steam and steam to liquid.

• **Isentropic process**
Steam is expanded to do work which causes the blades of turbine turning and compressed to operating pressure.

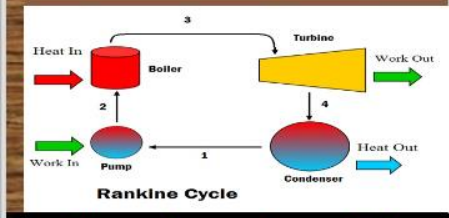
WORKING PRINCIPLE

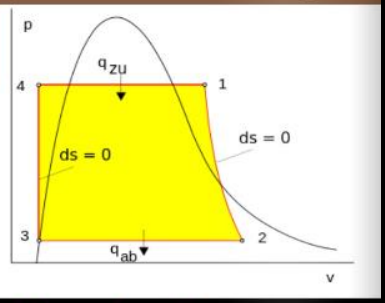
• **Isentropic compression**
1) Work is done to compress the working fluid by pump.
2) The pressure of water increases.

• **Isobaric expansion**
1) Heat is added to water through a fuel in boiler.
2) Water is converted into steam.

• **Isentropic expansion**
1) Steam expands and turns the blade of turbine.
2) Work is generated and converted into electrical energy.

• **Isobaric compression**
1) Heat is released from condenser to surrounding.
2) Steam is condensed into liquid.





Thermal power plant applies laws of thermodynamics to generate electricity.

The First law of Thermodynamics

The First law of Thermodynamics: special cases

Constant-volume process

$$\Delta E_{int} = Q$$

If the volume of a system is held constant, that system can do no work: $W = 0$

If heat is absorbed by a system ($Q > 0$), the internal energy increases

If heat is lost during the process ($Q < 0$), the internal energy decreases

The First law of Thermodynamics: special cases

Constant-volume process in real life?

1. Expulsion of heat in heat engines: waste heat is expelled right after the vehicle receives power from combustion, but since the piston is not moving, the volume of the chamber is unchanging and the pressure drops.
2. Boiling of water in a pressure cooker: When heat is transferred to the container, isovolumetric process is done since temperature and pressure increase is done at a constant volume.

The First law of Thermodynamics: special cases

Adiabatic process

$$\Delta E_{int} = -W$$

A process that occurs so rapidly or occurs in a system that is so well insulated that *no transfer of energy as heat* occurs between the system and its environment: $Q = 0$

If work is done *by* the system ($W > 0$), the internal energy decreases by the amount of work.

If work is done *on* the system ($W < 0$), the internal energy increases by by the amount of work.

The First law of Thermodynamics: special cases

Adiabatic process in real life?

1. Insulate the system: A vacuum flask does a good job of retaining heat - because of it's insulation.

2. Do not insulate the system:
An expanding and contracting sound wave (air compression and decompression) is a good example. It happens far too quickly for inflow or outflow of heat.

The First law of Thermodynamics: special cases

Cyclical process (Closed cycle)

$$Q = W$$

After certain interchanges of heat and work, the system is restored to its initial state: $\Delta E_{int} = 0$

The net work done during the process must exactly equal the net amount of energy transferred as heat

The store of internal energy of the system remains unchanged

The First law of Thermodynamics: special cases

Cyclical process in real life?

1. All the real cyclics behave as non-cyclical process, but…

2. A cyclic process is the underlying principle for an engine.

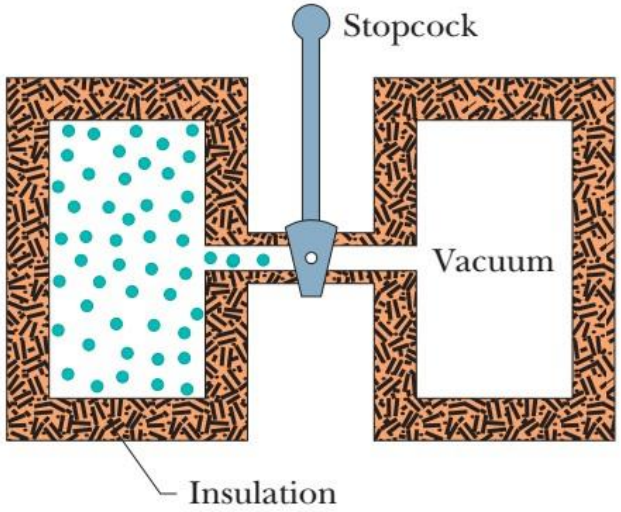
The First law of Thermodynamics: special cases

Free expanssions

$$\Delta E_{int} = 0$$

Adiabatic processes in which no transfer of heat occurs between the system and its environment and no work is done on or by the system:

$$Q = W = 0$$



The First law of Thermodynamics

The Law: $\Delta E_{\text{int}} = Q - W$

Process	Restriction	Consequence
Adiabatic	$Q = 0$	$\Delta E_{\text{int}} = -W$
Constant volume	$W = 0$	$\Delta E_{\text{int}} = Q$
Closed cycle	$\Delta E_{\text{int}} = 0$	$Q = W$
Free expansion	$Q = W = 0$	$\Delta E_{\text{int}} = 0$

The First law of Thermodynamics

1. $\Delta E_{int} = 0$ means that the process is isothermic. True or False?

a) True

b) False

Explanation: Internal energy is a function of temperature only. So if $\Delta E_{int} = 0$ implies that the process will have constant temperature, or it is isothermic.

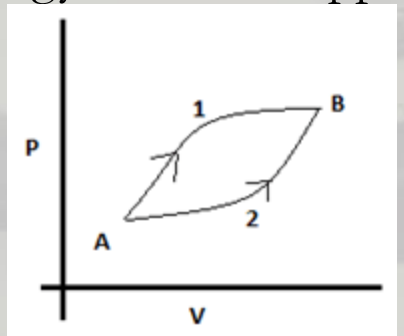
2. Relation between the internal energy and heat supplied in the process 1 & 2?

a) $\Delta E_{in1} > \Delta E_{in2}$, $Q_1 > Q_2$

b) $\Delta E_{in1} < U_2$, $Q_1 > Q_2$

c) $\Delta E_{in1} = \Delta E_{in2}$, $\Delta E_{in1} = \Delta E_{in2}$

d) $\Delta E_{in1} = \Delta E_{in2}$, $Q_1 > Q_2$



Explanation: The initial and final states are the same for both processes: $\Delta E_{int1} = \Delta E_{int2}$

The area under 1 (W_1) is greater than area under 2 (W_2)

$Q = \Delta E_{in1} + W \rightarrow Q_1 > Q_2$ since ΔE_{int} is same for both and $W_1 > W_2$

The First law of Thermodynamics

3. True or False?

a) $\Delta E_{int} = -W$ in an isothermal process

b) $\Delta E_{int} = W$ in an isothermal process

c) $\Delta E_{int} = -W$ in an adiabatic process

d) $\Delta E_{int} = -W$ in an adiabatic process

Answer: c

4. A system is taken from state A to state B along two different paths, 1 and 2, then:

a) $Q1 = Q2$

b) $W1 = W2$

c) $Q1 - W1 = Q2 - W2$

d) $Q1 + W1 = Q2 + W2$

Answer: c



Exercise

- Let 1.00 kg of liquid water at 100°C be converted to steam at 100°C by boiling at standard atmospheric pressure.
- The volume of that water changes from an initial value of 1.00×10^{-3} m^3 as a liquid to 1.671 m^3 as steam.

(a) How much work is done by the system during this process?

(b) How much energy is transferred as heat during the process?

(c) What is the change in the system's internal energy during the process?

Heat transfer mechanism

Conduction rate:

$$P_{cond} = \frac{Q}{t} = kA\,\frac{T_H - T_C}{L}$$

k: *thermal conductivity*

Conduction rate through composite:

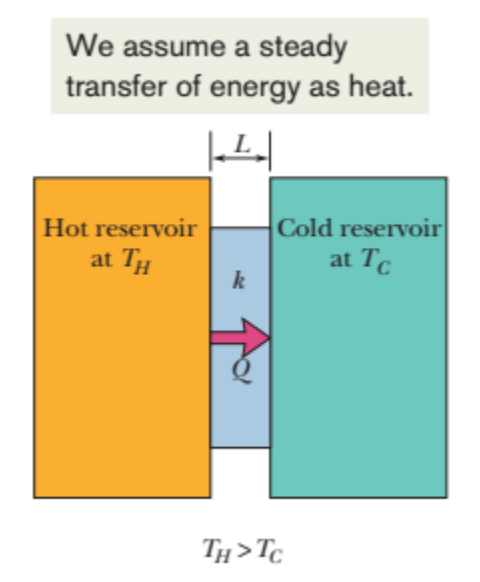
$$P_{cond} = \frac{A(T_H - T_C)}{\Sigma(\frac{L}{k})}$$

Thermal resistance:

$$R = \frac{L}{k}$$

High R-value: poor thermal conductor (good thermal insulator)

Consider a slab of face area A and thickness L, whose faces are maintained at temperatures T_H and T_C by a hot reservoir and a cold reservoir



Substance	k (W/m·K)
Metals	
Stainless steel	14
Lead	35
Iron	67
Brass	109
Aluminum	235
Copper	401
Silver	428
Gases	
Air (dry)	0.026
Helium	0.15
Hydrogen	0.18
Building Materials	
Polyurethane foam	0.024
Rock wool	0.043
Fiberglass	0.048
White pine	0.11
Window glass	1.0

Heat transfer mechanism

Rate at which an object emits energy via electromagnetic radiation:

$$P_{rad} = \delta \varepsilon A T^4$$

Rate at which an object absorbs energy via electromagnetic radiation:

$$P_{rad} = \delta \varepsilon A T^4$$

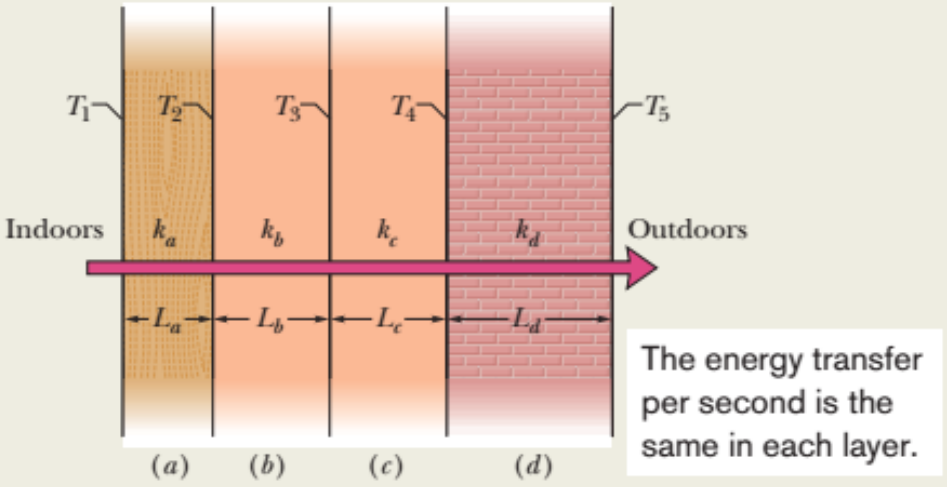
$$P_{net} = P_{abs} - P_{rad} = \delta \varepsilon A ({T_{env}}^4 - T^4)$$

$\delta = 5.6704 \times 10^{-8}$ W/m^2.K^4 (Stefan–Boltzmann constant)

ε: Emissivity (between 0 and 1)

Heat transfer mechanism

The figure shows the cross section of a wall made of white pine of thickness L_a and brick of thickness L_d (= $2.0L_a$), sandwiching two layers of unknown material with identical thicknesses and thermal conductivities. The thermal conductivity of the pine is ka and that of the brick is k_d (= $5.0k_a$). The face area A of the wall is unknown. Thermal conduction through the wall has reached the steady state; the only known interface temperatures are T_1 = 25°C, T_2 = 20°C, and T_5 = -10°C. What is interface temperature T_4?



$$P_a = k_a A \frac{T_1 - T_2}{L_a} \quad \text{and} \quad P_d = k_d A \frac{T_4 - T_5}{L_d}.$$

Setting $P_a = P_d$ and solving for T_4 yield

$$T_4 = \frac{k_a L_d}{k_d L_a}(T_1 - T_2) + T_5.$$

Letting $L_d = 2.0L_a$ and $k_d = 5.0k_a$, and inserting the known temperatures, we find

$$T_4 = \frac{k_a(2.0L_a)}{(5.0k_a)L_a}(25°C - 20°C) + (-10°C)$$

$$= -8.0°C. \qquad \text{(Answer)}$$